Paleomagnetism of the Nuanetsi Igneous Province and Its Bearing upon the Sequence of Karroo Igneous Activity in Southern Africa

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Further sites have been collected in the Marangudzi ring complex and the results confirm the presence of one normally magnetized site among seven reversely magnetized sites. A new paleomagnetic pole position is given for Marangudzi at 70.2°S, 105.1°E, with $A_{95} = 8.9°$. This pole is 15.6° away from the pole previously found for the nearby ring complexes at the Mateke Hills (58.6°S, 79.7°E, $A_{95} = 8.3°$). The difference is shown to be significant, and the two poles must therefore be quoted separately. The difference in pole position probably corresponds to a difference in age which recent radiometric age determinations reveal. A limited collection in the Nuanetsi lavas showed reversed magnetization in the first 2000 meters of the sequence, and this is correlated with the reversed magnetization found at Marangudzi. Data on ages and polarities for the whole of southern Africa are summarized, and a simple reversal sequence is suggested. A comparison is made with results from rocks of a similar age in other continents.

INTRODUCTION

In a previous investigation the paleomagnetism of the ring complexes at Marangudzi and the Mateke Hills in the southeast of Southern Rhodesia was studied [*Gough et al.*, 1964*a*]. The majority of sites showed reversed directions of magnetization, but two sites, one from Marangudzi and one from the Mateke Hills, possessed normal polarity, indicating that a field reversal may have occured during the emplacement of the complexes. It is known that mixed polarities occur in the Karroo dolerites of South Africa [*Graham and Hales*, 1957] and in the Stormberg lavas of Lesotho (formerly Basutoland). *van Zyl et al.* [1962*a*, 1962*b*] have documented a reversal in which the sequence is clearly from reversed to normal (R → N). The existence of a reversal in the Nuanetsi region is therefore of some importance for it may represent the same reversal as that recorded in the Stormberg lavas. In order to clarify the position it was necessary to obtain more information about the sequence of polarities in the Nuanetsi area, and further sampling was therefore planned.

GEOLOGY

The geology of the Nuanetsi igneous province is described by *Cox et al.* [1965] and a simplified map is shown in Figure 1. The Nuanetsi syncline is the main structural feature of the province: it has an east-west axis with an eastward plunge toward the end of which the volcanics reach a thickness of about 7500 meters. This structure allows the full sequence of lavas to appear at the surface, and depending on suitable outcrop, sampling is possible over the whole thickness. Postdating the lavas are a series of intrusive igneous bodies, which include a large granophyre sill and the three ring complexes, that were sampled in the earlier investigation. Since mixed polarities were observed in the ring complexes, the polarity of the lavas could have an important bearing upon the sequence of polarity during this period.

The geology of Marangudzi is shown in Figure 2, and is described in detail by *Gifford* [1961] and *Rees* [1960]. In outline it consists of an elongated mass of gabbro that contains small bodies of micro-gabbro and that is cut by two multiple ring intrusions, which consist of acid and undersaturated rocks, respectively. The site of normal polarity at Marangudzi was in the micro-gabbro, but reversed sites were found in both the main gabbro and the under-

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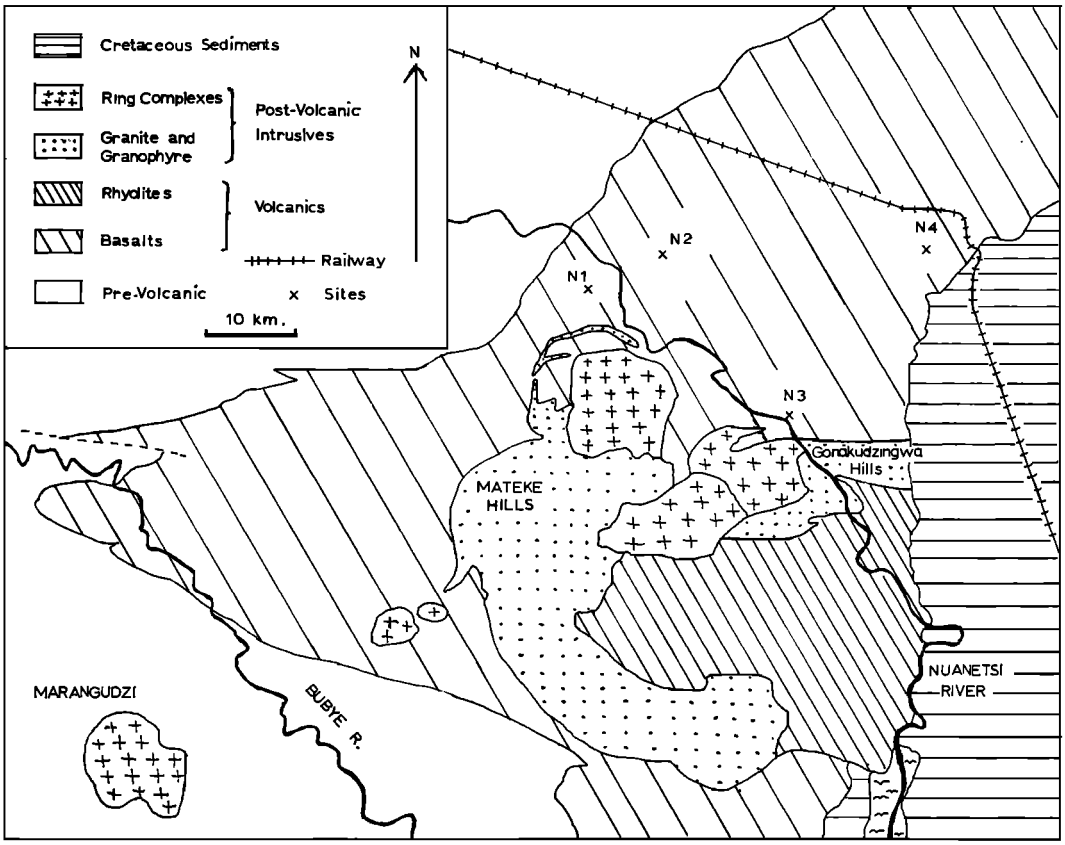


Fig. 1. Geological sketch map of the Nuanetsi area [after *Cox et al.*, 1964], showing the four sites sampled in the Nuanetsi lavas.

saturated ring. The polarity sequence at Marangudzi, therefore, depends upon whether the micro-gabbro was intruded toward the beginning or the end of the formation of the whole complex. In the earlier investigation the latter alternative was preferred, thus making the Marangudzi polarity sequence from reversed to normal (R → N) and creating the possibility that the reversal is the same as that in the Lesotho lavas. A. C. Gifford (private communication) has since pointed out that whereas micro-gabbros elsewhere in Marangudzi can clearly be seen to be intrusive into the main gabbro, the relationship of the main gabbro with the micro-gabbro that yielded the normal polarity site cannot be determined. Further sampling was, therefore, required at Marangudzi with particular reference to the micro-gabbro.

Field Work

A field trip was undertaken in February 1965, and a few reconnaissance samples had been collected when difficulties with the drilling equipment forced a return to Salisbury for repairs. A second trip was made in May, but on May 28 the Rhodesian authorities declared a state of emergency in the Nuanetsi district, the effect of which was to deny the author access to a large part of the area covered by the Nuanetsi lavas. Full sampling was, therefore, restricted to the Marangudzi complex, and the Nuanetsi lavas are represented by only four sites, at each of which only hand samples were obtained.

Samples were collected at Marangudzi at four sites (Figure 2). One (MH) was the original micro-gabbro site, two new micro-gabbro sites were sampled (MU and MX), and a site in the main gabbro (MA) was resampled, having yielded poor results in the earlier study. At all sites cores 60 cm long and 2.5 cm in diameter were obtained using the portable drilling equip-

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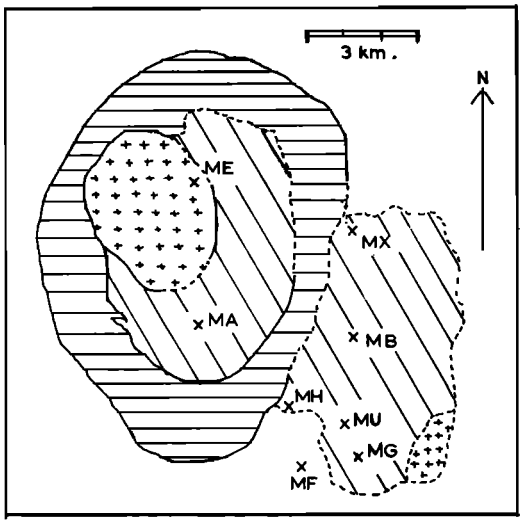


Fig. 2. Geological sketch map of the Marangudzi ring complex (after *Gifford,* [1961]) showing the eight sites given in Table 1. Diagonal lines indicate gabbro; horizontal lines, acid rocks; plus signs, undersaturated rocks; clear area, basement; crosses, site.

ment described by *McElhinny and Neal* [1967] The cores were oriented with respect to the sun's azimuth by the method of *Gough and Opdyke* [1963]. Table 1 gives details of these four sites and also shows the successful sites obtained in the first study.

In the Nuanetsi area only four sites were sampled (Figure 1). At each site from one to three blocks were collected after orientation with a Brunton compass-clinometer. The four sites are all from the first 2500 meters of the lava sequence, and the upper 5000 meters is, therefore, unrepresented. J. R. Vail has kindly estimated the positions of the four sites, and his figures are shown in Table 2. These estimates are approximate since details of faulting and of topography are not well known. The dip and strike were estimated from the map.

Measurements

Marangudzi. Cylindrical specimens 2.5 cm long were cut from the oriented cores, one from near the bottom and one at about 30 cm from the surface. These were regarded as independent samples in the statistical analysis. Directions and intensities of magnetization were measured with a spinner magnetometer [*Gough,* 1964], and partial alternating field demagnetization

TABLE 1. Sites at Marangudzi

Site Code (This Paper)	Previous Designation	Rock Type	Lat. S	Long. E
MA	A6	Gabbro	22°03′	30°39′
MH	A8	Micro-gabbro	22°05′	30°40′
MU		Micro-gabbro	22°01′	30°38′
MX		Micro-gabbro	22°02′	30°41′
MB	A7	Gabbro	22°04′	30°41′
ME	A3	Pulaskite	22°01′	30°39′
MF	A11	Lamprophyric dike	22°05′	30°40′
MG	A10	Gabbro	22°05′	30°41′

was carried out with the apparatus described by *McElhinny* [1966]. At all sites the initial remanent magnetization showed widely scattered directions, but partial demagnetization in fields ranging from 30 to 80 milliweber m^{-2} was successful in removing secondary components. Good well-grouped directions were measured after magnetic cleaning.

At Marangudzi the micro-gabbro sites MU and MX both gave well-grouped reversed directions. They were, therefore, in agreement with the other sites at Marangudzi but in disagreement with the original micro-gabbro MH. At site MH itself, four new samples gave normal directions of magnetization in agreement with the original nine. Because the area of outcrop at MH is comparatively small and because the new samples were taken within a few tens of meters of the original cores, it was decided not to create a second site for the analysis but to combine all the results to give a single new site of thirteen samples. At site MA magnetic cleaning produced a good group with a re-

TABLE 2. Stratigraphic Positions above the Base of the Sequence of the Four Sites from the Nuanetsi Lavas

Site Code	Number of Successful Samples	Estimated Height above Base, meters
N1	2	450
N2	1	420
N3	2	2200
N4	1	1500

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TABLE 3. Mean Directions of Magnetization and Site Pole Positions from the Eight Sites at Marangudzi after Alternating Field Demagnetization

Site Code	B, milliweber m^{-2}	N, Number of Samples	M					Site Pole		Polarity
			D	I	R	k	α_{95}	Lat. S	Long. E	
MA	80	6	178.7	+30.3	5.903	51.8	9.4	84.1	198.6	R
MH	40–80	13	340.6	−35.2	12.704	40.5	6.4	71.6	125.6	N
MU	30	6	168.2	+35.8	5.925	66.9	8.3	78.7	129.7	R
MX	40	6	151.0	+47.9	5.966	146	5.6	63.0	99.7	R
MB	30	12	157.6	+43.2	11.949	216	3.0	69.3	107.9	R
ME	30	11	139.7	+43.0	10.928	140	3.9	53.1	108.1	R
MF	60	9	156.1	+48.3	8.763	33.7	9.0	67.3	96.9	R
MG	60–80	9	155.1	+58.3	8.683	25.2	10.4	62.8	76.3	R

Note.

M, Remanent magnetization after demagnetization in alternating field of peak value B.
D, declination of remanent magnetization.
I, inclination of remanent magnetization.
R, length of the resultant of N vectors.
k, $(N - 1)/(N - R)$ = estimate of Fisher's κ parameter.
α_{95}, semi-angle of cone of 95% confidence for the mean direction.

versed direction. In the original investigation magnetic cleaning in fields up to the maximum then available (80 milliweber m^{-2}) was unsuccessful with specimens from the short 15-cm cores. The longer 60-cm cores drilled at the same site and treated in similar fields now give a result that is in agreement with the other reversed sites at Marangudzi. This illustrates the improvement that can be obtained by sampling at greater depths. Table 3 gives details of the four new sites under discussion and also repeats the data for sites MB, ME, MF, and MG from the earlier study. The number of successful sites at Marangudzi is thus increased from five to eight, and the new mean direction for Marangudzi is shown in Table 4. The mean direction is calculated both for the seven reversed sites and for all eight sites with an assumed reversal for the one normal site. The 'reversed' normal site falls well within the group formed by the other seven sites, and there would be no reason to exclude it from the mean except for the reversal. Accordingly, the preferred mean direction for Marangudzi is taken as that which includes all eight sites.

TABLE 4. Mean Directions

Rock Unit	N	D	I	k	α_{95}
Marangudzi (all sites)	8	159.3	+43.6	41.3	8.7
Marangudzi (normal site excluded)	7	159.0	+44.8	38.2	9.9
Nuanetsi (six samples)	6	110.3	+71.7	8.08	25

N, D, I, k, and α_{95} as in Table 3.

Nuanetsi. A total of ten oriented blocks were collected from four sites in the Nuanetsi lavas. From the blocks two cores 2.5 cm in diameter were drilled in the laboratory, and one cylinder 2.5 cm in length was cut from each core. The mean of the two specimens from each block was then treated as a single observation for the analysis. The ten samples responded poorly to the magnetic cleaning technique, and no stable end point could be found in four cases. The remaining six samples formed a rather loose group after demagnetization in fields between 80 and 100 milliweber m^{-2}. All six were reversed in polarity and Table 4 shows their mean directions. It was not thought justified to analyze these samples in terms of collection sites, because two of the sites are represented by only one sample.

Discussion of Results

Pole positions. Paleomagnetic pole positions

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TABLE 5. Mean Paleomagnetic Poles in Southern Hemisphere

Rock Unit	Pole Positions	Lat. S	Long. E	Number of Sites	K	A_{95}	δp	δm
Marangudzi	G	70.2	105.1	8	39.9	8.9		
		70.7	106.7				6.8	10.9
Mateke Hills	H	58.6	79.7	6	65.8	8.3		
[*Gough et al.*, 1964*a*]		58.7	80.5				7.0	9.4
Marangudzi and Mateke Hills [*Gough et al.*, 1964*a*]	GH	60.9	86.2	10	57	6.5		

K, estimate of Fisher's precision parameter.
A_{95}, semiangle of cone of 95% confidence for the mean pole (calculated from site poles).
δp and δm, are the semi-axes of the ellipse of 95% confidence for the pole (calculated from the mean of directions).

are given in Table 5. For Marangudzi the pole is calculated in two ways; the first involves the computation of a mean pole from the eight site poles, and the second yields a pole from the mean of the eight site directions. The difference between the poles calculated by the two methods is small, but both are shown for reference.

Table 5 also includes the pole position for the six reversed sites from the Mateke Hills as reported by *Gough et al.* [1964*a*]. The seventh site at Mateke Hills had normal polarity and has significance in relation to the possible sequence of polarities, but, because its direction after allowing for the reversal lay some 35° away from the remainder of the group, it was not included in the calculation.

In the original investigation the pole positions from Marangudzi and the Mateke Hills were first reported separately. It was then shown that the pole positions were not significantly different at the 95% level, and a final pole position was quoted for the combination of Marangudzi and the Mateke Hills. This pole is also shown in Table 5. The combination of the two was supported by the presence of mixed polarity in both intrusions and by their geological similarity, which jointly suggested that Marangudzi and the Mateke Hills intrusions were approximately contemporaneous.

When the new Marangudzi pole is compared with the Mateke Hills pole, it is found that the pole positions are now 15.6° apart, which makes them significantly different at the 95% confidence level. The procedure is the same as that outlined by *Jones and McElhinny* [1966] whereby it is first necessary to test for identity of dispersions of the two populations before going on to apply the test for identity of polar vectors [*Watson and Irving*, 1957]. The two poles and their circles of error are shown in Figure 3. It can be seen that the circles of confidence overlap slightly, and it is, therefore, not possible to apply the simple rule that mean polar directions are significantly different when their circles of confidence do not overlap. In this case, however, the overlap is not large, and a full statistical calculation gives the result already mentioned.

If Marangudzi and the Mateke Hills are contemporaneous, it is difficult to account for the

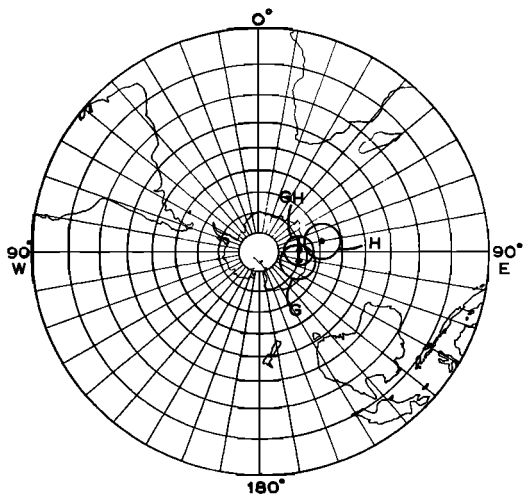


Fig. 3. The three pole positions summarized in Table 5.

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difference between the pole positions except by assuming that secular variation has not been averaged out in the sampling. But the number and spread of sites in both cases are reasonable, and the values of angular dispersion are consistent with those for present-day field variations [*McElhinny et al.*, 1968], suggesting that secular variation has been eliminated. The alternative is that there is a real difference in age between Marangudzi and the Mateke Hills, such that a certain amount of polar wander occurred between their emplacements.

Radiometric age determinations by the rubidium-strontium method have recently been carried out on rocks from the Mateke Hills and the Nuanetsi lavas by W. I. Manton, who has kindly supplied the figures given in Table 6. W. I. Manton used the decay constant $\lambda = 1.39 \times 10^{-11}$ yr^{-1} and quoted errors at the 90% confidence level. The table also shows the ages according to the alternative decay constant $\lambda = 1.475 \times 10^{-11}$ yr^{-1} because the latter is often preferred when comparing with ages obtained by the potassium-argon method. The age of Marangudzi has already been determined by J. A. Miller at about 190 m.y. and was quoted by *Gough et al.* [1964*a*]. This is in agreement with W. I. Manton's figures, but his age for two of the Mateke Hills ring complexes is significantly younger than that for Marangudzi. It now appears that there is a difference in age of the order of 20 m.y. between the two sets of intrusions. It is, therefore, reasonable to suggest that the difference in pole position reflects this age difference. The implication is that a certain amount of polar wander did occur during this part of the Mesozoic. It does appear, however, that this movement took the form of small excursions of the order of 10° to 15° about a position that remained approximately constant relative to Africa for most of the Mesozoic era [*Gough et al.*, 1964*b*].

TABLE 6. Ages of Samples from the Nuanetsi Province

Rock Unit	Age, m.y.
Granites from Masukwe and Dembe–Divula ring complexes	177 ± 4 (168)
Rhyolites from the Nuanetsi district	206 ± 13 (194)
Rhyolites from Swaziland and Northern Zululand (Lebombo)	202 ± 14 (190)

Note$_n$ The ages in parentheses are those corresponding to the decay constant $\lambda = 1.475 \times 10^{-11}$ yr^{-1}.

The paleomagnetic pole for the Nuanetsi lavas is not quoted. The pole has a very large error associated with it, and this, together with the small size of the sample and the weathered character of the rocks, makes it extremely unwise to use the result as an estimate of the paleomagnetic pole for the lavas. It is, however, certain that the four lava sites have reversed polarity rather than normal polarity, and it is this fact that has significance for the rest of the discussion.

Reversals. The results obtained here can also be used in connection with the problem of reversals. The new results confirm that only one site at Marangudzi is normal in polarity, and it establishes the existence of reversed polarity in the first 2500 meters of the Nuanetsi lava sequence.

The question of whether the change of polarity observed at Marangudzi is from normal to reversed or from reversed to normal has already been seen to depend upon whether the micro-gabbro at the one 'normal' site was early or late, respectively, in the formation of Marangudzi. The results show that the two other sites in micro-gabbro have reversed polarity in agreement with the main gabbro and the undersaturated ring. These reversed micro-gabbros and the pulaskite site from the undersaturated ring postdate the main gabbro, and, if the polarity sequence is to be R → N, then the normal micro-gabbro must be a very late stage of the intrusion. A. C. Gifford has said that this is difficult to accept and he prefers the suggestion that the normal micro-gabbro may be an early stage chilled margin, in support of which he points out that this particular micro-gabbro shows strong flow banding. The contact is not exposed, but the map shows that the normal micro-gabbro site (MH) is closer to the presumed contact than any of the other sites. If these considerations are accepted and if the site MH is assumed to be in a chilled margin of Marangudzi, then the polarity change at Marangudzi must be from normal to reversed, and

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it cannot be the same as that recorded in the Lesotho lavas.

There is a possibility that the reversal at site MH is an example of self–reversal. A simple test of this possibility was made by the original authors with a negative result. The same test was applied to another eight specimens obtained in this study. Four specimens from the normal micro-gabbro and two each from the reversed micro-gabbro sites MU and MX were heated to 720°C and allowed to cool in an applied field of the same order of magnitude as the geomagnetic field. All specimens acquired normal magnetization on cooling. Although this is not conclusive, it does suggest that site MH records a genuine field reversal rather than being an example of self-reversal.

The sequence of reversals. If the results from other African rocks of Triassic-Jurassic age are combined with those reported here, it becomes possible to suggest a sequence of polarities for this period of geologic time. In doing this it is necessary to compare radiometric ages by both the Rb–Sr and the K–Ar methods. Following the general practice [*Harland et al.*, 1964], all Rb–Sr ages are converted to the decay constant $\lambda = 1.475 \times 10^{-11}$ yr^{-1} for this comparison.

During most of the Middle and Upper Triassic the earth's field had normal polarity as recorded in the Triassic redbeds of Zambia [*Opdyke*, 1964] and the Cave sandstone [*Nairn*, 1960] and at Shawa [*Gough and Brock*, 1964], which is dated at 197 ± 15 m.y. Near the Triassic-Jurassic boundary a change from normal to reversed polarity took place and is recorded at Marangudzi, where a chilled margin seems to have 'caught' the end of the normal period. The first 2000 meters of the Nuanetsi lava sequence show reversed polarity, and the upper part of the sequence, which includes the rhyolites, is dated at 194 ± 12 m.y. Marangudzi has an age of about 190 m.y. with individual samples ranging from 182 m.y. to 196 m.y. If the reversal is put at about 195 m.y., it is not inconsistent with the ages given here. At the Mateke Hills, which intrude the Nuanetsi lavas, six sites are reversed and one site is normal. The simplest hypothesis is that the period of reversed polarity, whose beginning is recorded at Marangudzi and in the Nuanetsi lavas, is also shown at the six Mateke Hills sites; the seventh site, which has a normal polarity, represents the next normal period. The Mateke Hills may, therefore, date the end of the reversed period, which would then be at about 170 m.y. Since the Lesotho lavas show the same change from reversed to normal polarity, it is tempting to suggest that they are of a similar age and record the same reversal.

Lavas of Stormberg age elsewhere in Rhodesia have been measured by *Nairn* [1960] and by *Opdyke* [1967]. The results show that the lava flows at Victoria Falls are of normal polarity. Several sites in the Bulawayo and Gokwe areas are also of normal polarity, and one site at Felix Farm in flow BK1 shows an intermediate direction. It is, therefore, suggested that these lavas were extruded during the normal polarity period that followed the intrusion of the Mateke Hills and that is recorded in the upper 780 meters of the Lesotho lava sequence.

The suggested sequence of events is shown in Table 7. The sequence is a simple one and was suggested by *Opdyke and McElhinny* [1965] before the Nuanetsi results became available.

If the sequence can be firmly established, it means that, during this part of the Mesozoic, a relatively long period of reversed polarity occurred with relatively long periods of normal polarity predating and postdating it. *Irving* [1964] has remarked that this provides the most favorable situation for the use of reversals as worldwide marker horizons, and the possibility exists that these reversals can provide a useful stratigraphic tool for long distance correlation.

TABLE 7. Sequence of Polarities in Africa

Formation	Age, m.y.	Polarity
Triassic red beds		N
Cave sandstone		N
Shawa	197 ± 15	N
Marangudzi	196–182	N → R
Nuanetsi lavas	194 ± 12	R
Mateke Hills	168 ± 4	R (N at end?)
Lesotho lavas		R → N
Batoka basalts		N (R at base?)
Stormberg dolerites	154–190 [*McDougall*, 1963]	N + R

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TABLE 8. Polarities outside Africa

Formation	Age, m.y.	Polarity	Reference
Nova Scotia dyke, N. America	197 ± 32	N	*Larochelle and Wanless* [1966]
Pallisades sill, N. J., N. America	193 ± 3	N	*Opdyke* [1961]
White Mountain series, New Hampshire, N. America	180	N + R	*Opdyke and Wensink* [1966]
Gibralter syenite, Australia	178	N	*Irving* [1964]
Prospect dolerite, Australia	168	N	*Irving* [1964]
Tasmanian dolerite, Australia	159–168	N	*Irving* [1964]
Ferrar dolerites, Antarctica	147–163	N	*Irving* [1964]
Serra-Geral formation, S. America	124–149	N + R	*Creer et al.* [1965]

A further consequence of the suggested sequence is that there must have been at least two distinct periods of lava outflow within the whole Stormberg volcanic sequence. The first resulted in the Nuanetsi and Lebombo lavas and coincided with the emplacement of the Marangudzi ring complex, and the second occurred some 25 m.y. years later and gave rise to the Lesotho and Victoria Falls lavas.

It is known that the frequency of reversals has varied in geological time. The past 25 m.y. has shown many reversals, and, if a similar frequency of reversal were present during the Mesozoic, it would probably be impossible to establish a unique polarity sequence. Although it is unlikely that such a situation existed, it is not impossible that a rather more complex sequence involving two or, possibly, three reversed periods could have existed between about 200 and 160 m.y. The African results, alone, suggest one reversed period, but it is necessary to review results from other parts of the world before constructing a definitive sequence. One can use results only from the rocks that have received both paleomagnetic and radiometric age studies. Table 8 lists the presently available results. The figures here are consistent with a period of reversed polarity in the Jurassic-Triassic and would allow the change N → R to be placed at about 195 m.y. The African figures would place the corresponding R → N change at about 170 m.y., but results from both Australia and North America demand normal polarity at about 180 m.y. It is possible that a normal event occurred at about this time, and further evidence may destroy the simple sequence of polarities suggested in Table 7. Further work is clearly required, but the reversals at about 195 m.y. and 170 m.y. do seem to be well established and may still provide a useful correlation tool. A further reversal at about 140 m.y. is suggested by evidence from South America [*Creer*, 1962; *Creer et al.*, 1965].

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